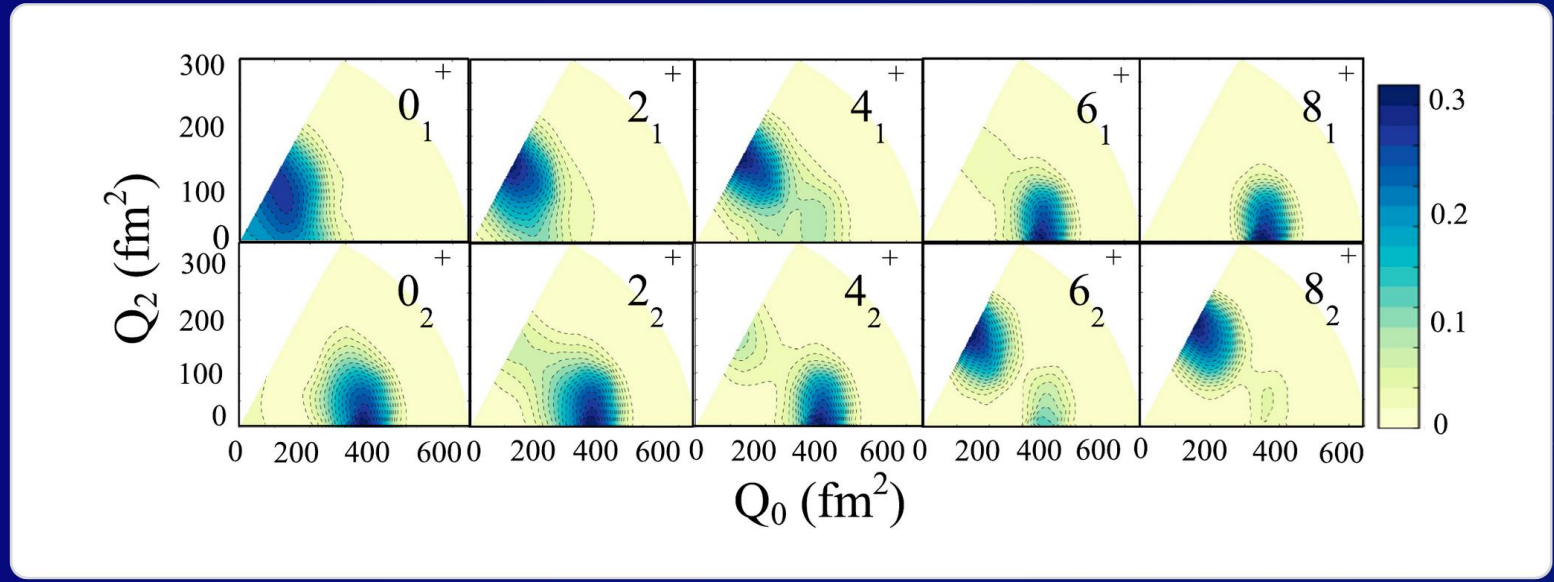
Shape coexistence and intruder-normal band crossing in even-even Sn isotopes

K. Kaneko, T. Mizusaki, N. Shimizu, and Y. Sun

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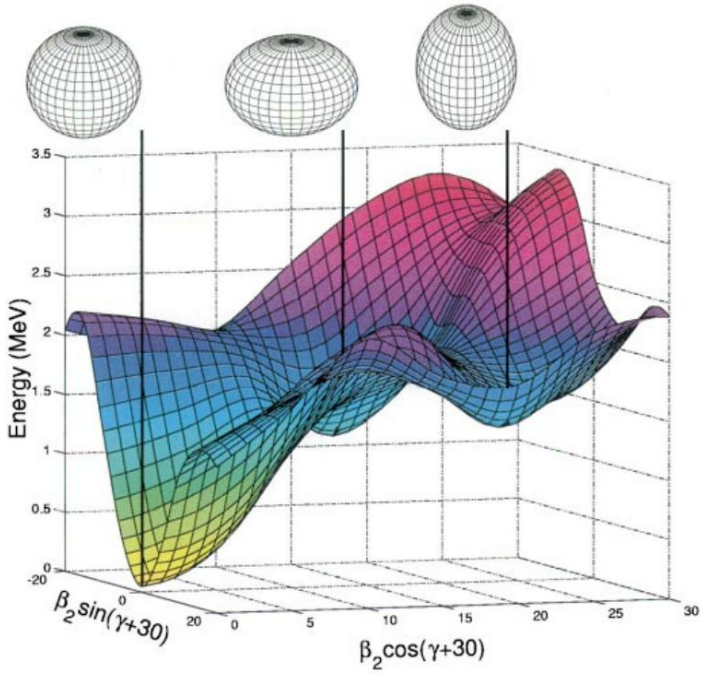


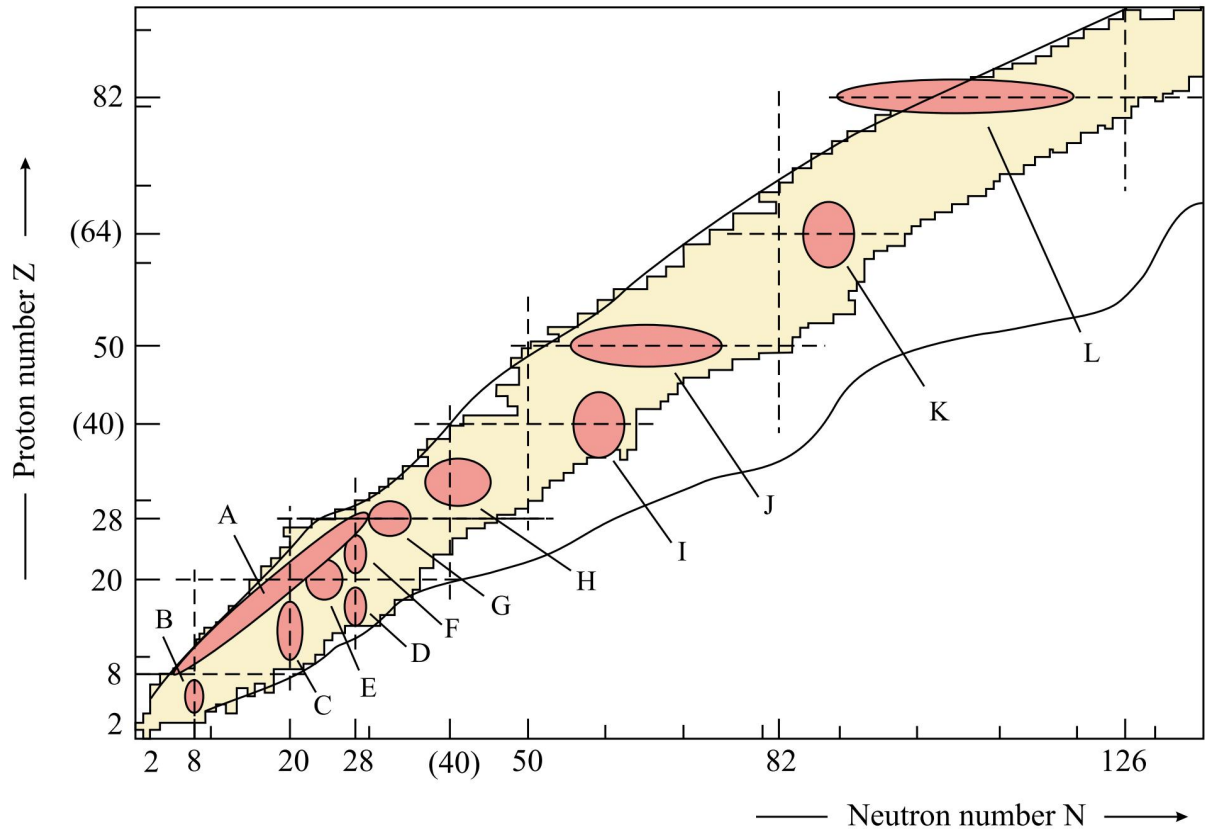
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Shape coexistence and its microscopic origin

Low-lying states with different intrinsic deformations

- Competing configurations may occur within a small excitation-energy interval. Cross-shell promotion costs shell-gap energy, while pairing, monopole and quadrupole correlations lower the intruder configuration and can produce a midshell minimum.



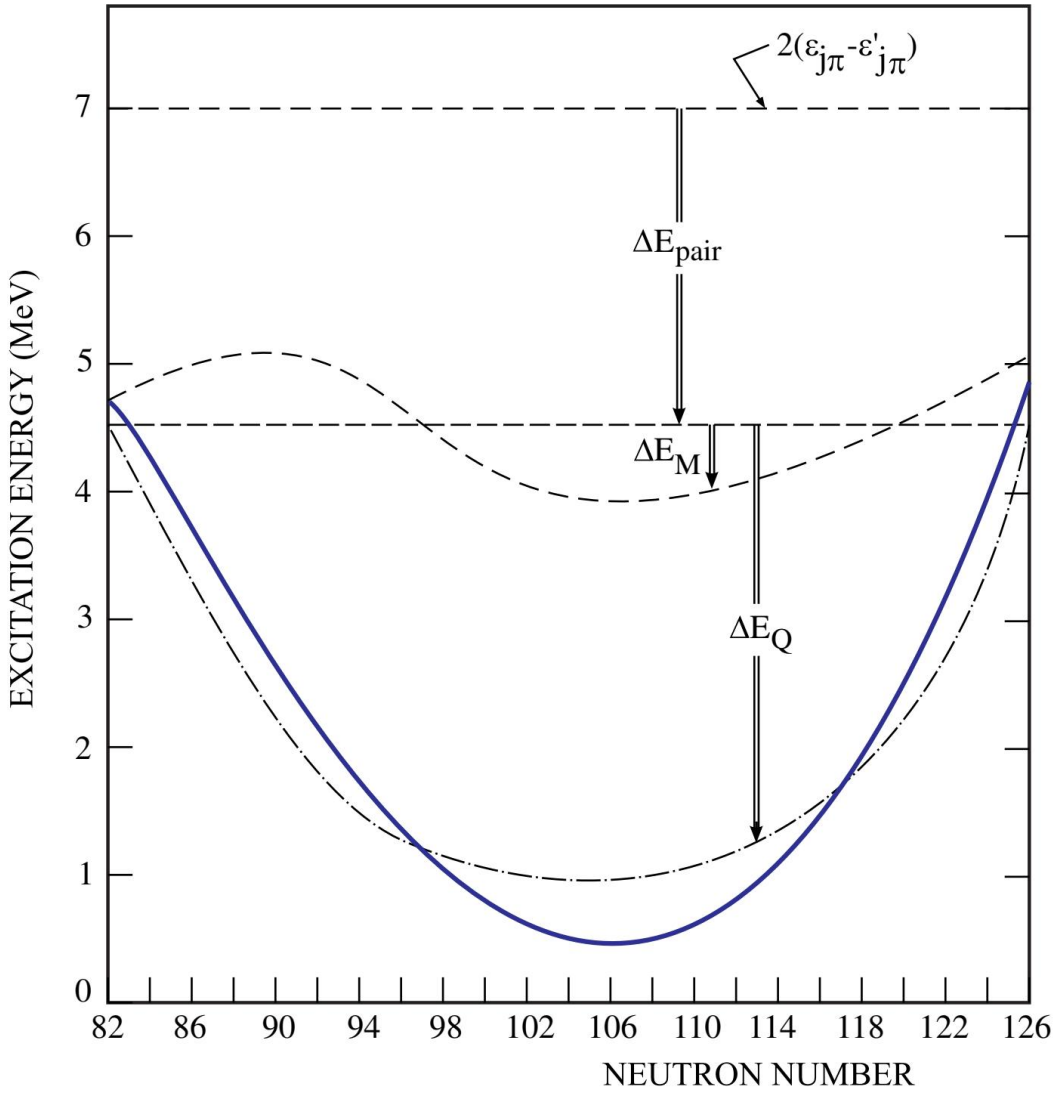


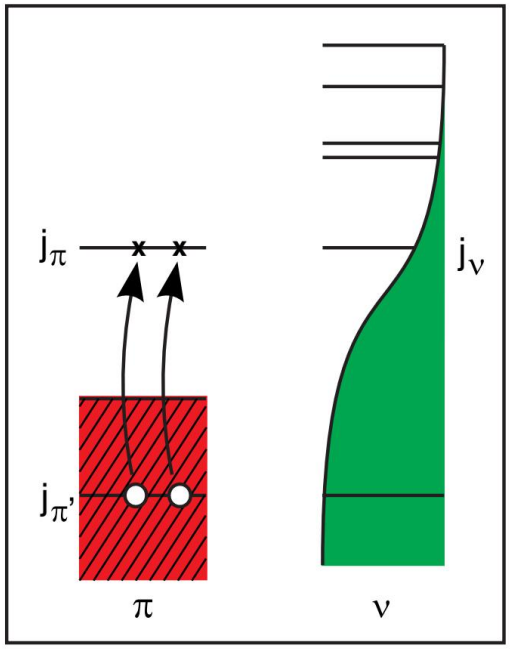
A. N. Andreyev *et al.*, Nature 405, 430 (2000)

K. Heyde and J. L. Wood, RMP 83, 1467 (2011)

Shape coexistence and its microscopic origin

Low-lying states with different intrinsic deformations





The parabolic lowering of the proton 2*p*-2*h* intruder state toward mid-shell originates mainly from the enhanced proton-neutron quadrupole correlation. Pairing provides an overall energy gain, whereas the monopole interaction gives a smaller, smoothly varying correction.

K. Heyde and J. L. Wood, RMP 83, 1467 (2011)

Shape coexistence in Sn before the present work

Experimental evidence, theoretical descriptions and open issues

Experiment

- Brön *et al.*: enhanced (^{3}He,n) population of excited 0^{+} states in $^{112\text{-}118}$Sn; the states were interpreted as proton 2p–2h intruder bandheads across $Z = 50$.
- Recent spectroscopy: strong normal–intruder mixing in ^{114}Sn and ^{116}Sn; ^{118}Sn appears less mixed. The intruder 0^{+} assignment in ^{116}Sn is not settled.

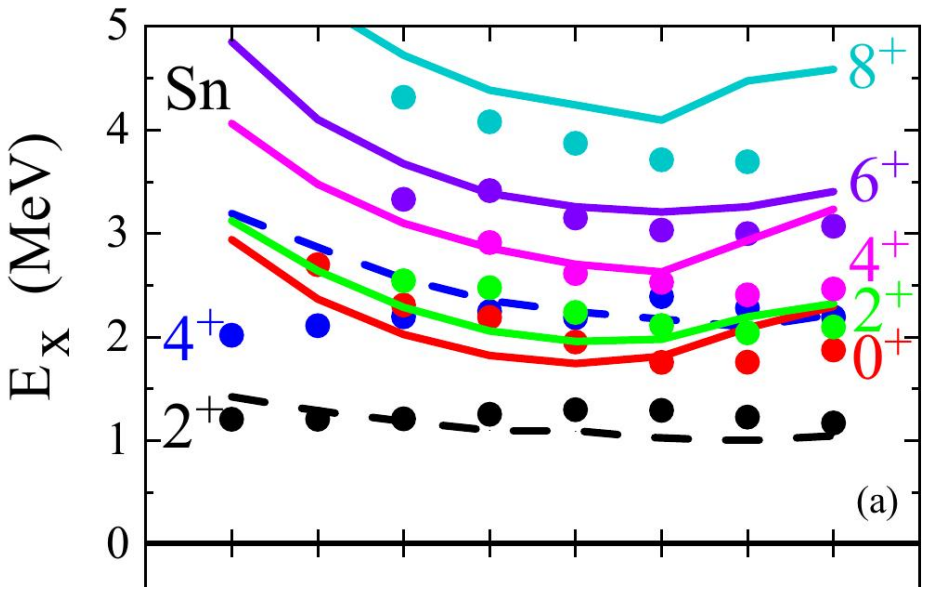
Theory

- *sd* IBM-2 with core breaking describes both structures phenomenologically, but assumes the 2p–2h proton configuration.
- LSSM spaces underestimate $B(E2; 0^{+} \rightarrow 2^{+})$ in the lower half of the shell. $\pi(g, d, s)$ and $\nu(g_{7/2}, d, s, h_{11/2})$
- Large-space HFB+GCM and MCSM identify $\pi 1g_{9/2} \rightarrow \pi 2d_{5/2}$ quasi-SU(3) excitations as important for the enhanced *E*2 strength.
- Open issue: a microscopic, equal-footing description of both bands, the number of promoted protons, and the $N = 66$ minimum of the intruder sequence.

Scope of the present Letter

Large-space shell model: normal and intruder states on equal footing

- PMMU Hamiltonian in the *gdshf* space, solved by particle-number- and angular-momentum-projected HFB+GCM for $^{106\text{-}120}$Sn The same wave functions are used for spectra, occupations, E2 strengths, Q_s, g factors, PESs and collective wave function distributions.



Shape coexistence and intruder-normal band crossing in even-even Sn isotopes

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We investigate microscopically systematic behavior of shape coexistence in even-even $^{106-120}$Sn isotopes in the framework of the shell model. The two-particle and two-hole (2p-2h) proton excitations across the $Z = 50$ closed shell gap are essential to describe the deformed intruder states built on the first excited 0_2^+ state, which intrudes into the spherical normal states, indicating the shape coexistence. In addition, the neutron excitations to the $1h_{11/2}$ orbit are important for a parabolic behavior of the intruder band with a minimum in the excitation energy near the neutron midshell $N = 66$ point. The calculations predict that the crossing between the intruder and normal bands occurs between spin $I = 4$ and $I = 6$. From the potential energy surface (PES) with particle-number and angular-momentum projections, collective wave function, and quadrupole moments, it is shown that the calculations describe shape coexistence structure for the midshell Sn nuclei.

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Questions addressed in this Letter

The two questions stated in the Introduction

- How many protons are excited across the $Z = 50$ shell gap?
- Why do the intruder-state energies have a minimum at $N = 66$ and exhibit parabolic trends?

The calculation then follows the normal and intruder sequences with spin in ^{116}Sn: where do they cross, how does the yrast shape change, and do occupations, Q_s, projected PESs and collective wave functions support the same band assignment?

PMMU shell-model Hamiltonian

P + QQ interaction with a monopole term

The Hamiltonian used in the Letter is defined as

$$H = H_0 + V_{P_0} + V_{P_2} + V_{QQ} + V_m^{MU}, \qquad (1)$$

H_0: single-particle term. v_{P0}: monopole pairing. v_{P2}: quadrupole pairing. V_{QQ}: quadrupole–quadrupole interaction.

The first four terms are usually referred as P+QQ Hamiltonian, assuming the form

$$\hat{H} = \hat{H}_0 - \frac{\chi}{2}\sum_{\mu}\hat{Q}_{\mu}^{\dagger}\hat{Q}_{\mu} - G_M\hat{P}^{\dagger}\hat{P} - G_Q\sum_{\mu}\hat{P}_{\mu}^{\dagger}\hat{P}_{\mu}.$$

The one-body operators above are defined by

$$\hat{Q}_{\mu} = \sum_{\alpha\beta} c_{\alpha}^{\dagger} Q_{\mu\alpha\beta} c_{\beta}\,, \quad \hat{P}^{\dagger} = \frac{1}{2}\sum_{\alpha} c_{\alpha}^{\dagger} c_{\bar{\alpha}}^{\dagger}\,, \quad \hat{P}_{\mu}^{\dagger} = \frac{1}{2}\sum_{\alpha\beta} c_{\alpha}^{\dagger} Q_{\mu\alpha\beta} c_{\bar{\beta}}^{\dagger}$$

where $\bar{\alpha}$ represents the time reversal of α, and $Q_{\mu\alpha\alpha'} = \delta_{NN'}\,(Njm\,|\,Q_{\mu}\,|\,N'j'm')$

PMMU shell-model Hamiltonian

P + QQ interaction with a monopole term

For monopole interactions, the monopole matrix elements are defined as

$$V_m(ab,T) = \frac{\sum_J (2J+1) V^{JT}_{ab,ab}}{\sum_J (2J+1)},$$

and the monopole interaction can be written as

$$H_m = \sum_{a \leqslant b,T} V_m(ab,T) \sum_{JMK} A^{\dagger}_{JMTK}(ab) A_{JMTK}(ab),$$

Otsuka et al. suggest "monopole-based universal force" (V_{MU})

$$V_{MU} = V_{\text{central}} + V_{\text{tensor}},$$

$$V_{\text{central}} = \sum_{S,T} f_{S,T} P_{S,T} \exp[-(r/\mu)^2], \quad V_{\text{tensor}} = (\vec{\tau}_1 \cdot \vec{\tau}_2)([\vec{s}_1 \vec{s}_2]^{(2)} \cdot Y^{(2)}) g(r),$$

where *S* (*T*) means spin (isospin), P denotes the projection operator onto the (*S*, *T*) channels, with strength f, internucleon distance r, and Gaussian parameter *μ*.

PMMU shell-model Hamiltonian

P + QQ interaction with a monopole term

The π + ρ meson exchange force is used for the strength $g(r)$

and the monopole interaction can be rewritten in the following form:

$$H_m = \sum_{a \leqslant b} \left[r_{ab} \hat{n}_a (\hat{n}_b - \delta_{ab}) + s_{ab} \left(\hat{T}_a \cdot \hat{T}_b - \frac{3}{4} \hat{n}_a \delta_{ab} \right) \right],$$

with

$$r_{ab} = \frac{3V_m(ab, T=1) + V_m(ab, T=0)}{4(1+\delta_{ab})},$$

$$s_{ab} = \frac{V_m(ab, T=1) - V_m(ab, T=0)}{1+\delta_{ab}}.$$

The monopole term controls effective single-particle energies and shell evolution.
Its matrix elements are slightly corrected and multiplied by $(102/A)^{0.3}$.

Model space: explicit cross-shell degrees of freedom

The seven-orbit gdshf space

- The space retains π1$g_{9/2}$ holes, the upper-shell proton partners required for cross-shell quadrupole correlations, and ν1$h_{11/2}$.

$$\mathcal{M}_{gdshf} = \{1g_{9/2},\ 1g_{7/2},\ 2d_{5/2},\ 2d_{3/2},\ 3s_{1/2},\ 1h_{11/2},\ 2f_{7/2}\}.$$

below Z (N) = 50 between 50 and 82 above N (Z) = 82

proton cross-shell degree of freedom $\pi 1g_{9/2} \rightarrow \pi 2d_{5/2} \quad (Z = 50),$

neutron correlation diagnostic lower orbits $\rightarrow \nu 1h_{11/2}.$

HFB + GCM for the shell-model Hamiltonian

A variational approximation to large-scale diagonalization

H is the PMMU valence-space Hamiltonian; HFB+GCM is the many-body solver used to approximate its eigenstates.

$$H_{\mathrm{PMMU}} \longrightarrow |\Phi(q)\rangle_{\mathrm{VAPNP}} \longrightarrow \hat{P}^I_{MK}\hat{P}^\pi\hat{P}^N\hat{P}^Z|\Phi(q)\rangle,$$

$$\xrightarrow{\mathrm{HWG}} |\Psi^{I\pi}_{\sigma M}\rangle \longrightarrow \{E,\, B(E2),\, Q_s,\, g,\, n_a\}.$$

VAPNP-HFB vacua describe pairing and deformation. Particle number, angular momentum and parity are projected, and the nonorthogonal basis is mixed in q and K through the Hill–Wheeler–Griffin equation. Benchmarks in this mass region give results of comparable quality to exact diagonalization and MCSM for the observables considered.

Projection, HWG mixing and collective amplitudes

Good N, Z, I and parity before configuration mixing

Projected GCM wave function

$$|\Psi^{I\pi}_{\sigma M}\rangle = \sum_{qK} f^{I\pi}_{\sigma K}(q)\, \hat{P}^{I}_{MK} \hat{P}^{\pi} \hat{P}^{N} \hat{P}^{Z} |\Phi(q)\rangle.$$

The wave function is determined by solving the Hill-Wheeler-Griffin equation

$$\sum_{q'K'} \left[\mathcal{H}^{I\pi}_{qK,q'K'} - E^{I\pi}_{\sigma} \mathcal{N}^{I\pi}_{qK,q'K'}\right] f^{I\pi}_{\sigma K'}(q') = 0,$$

where $\mathcal{H}$ and $\mathcal{N}$ are Hamiltonian kernels and norm kernels. The collective wavefunction amplitudes and distributions are

$$g^{I\pi}_{\sigma K}(q) = \sum_{q'K'} [\mathcal{N}^{I\pi}]^{1/2}_{qK,q'K'} f^{I\pi}_{\sigma K'}(q'), \qquad P^{I\pi}_{\sigma}(q) = \sum_{K} \left|g^{I\pi}_{\sigma K}(q)\right|^2.$$

Occupation-number indicators and observables

Configuration indicators

The paper defines proton holes in $\pi 1g_{9/2}$ and neutron promotion into $\nu 1h_{11/2}$ relative to the 0_1^+ state:

$$n_{\mathrm{ex}}(I_i^+, \pi 1g_{9/2}) = n(0_1^+, \pi 1g_{9/2}) - n(I_i^+, \pi 1g_{9/2}), \quad (2)$$

$$n_{\mathrm{ex}}(I_i^+, \nu 1h_{11/2}) = n(I_i^+, \nu 1h_{11/2}) - n(0_1^+, \nu 1h_{11/2}), \quad (3)$$

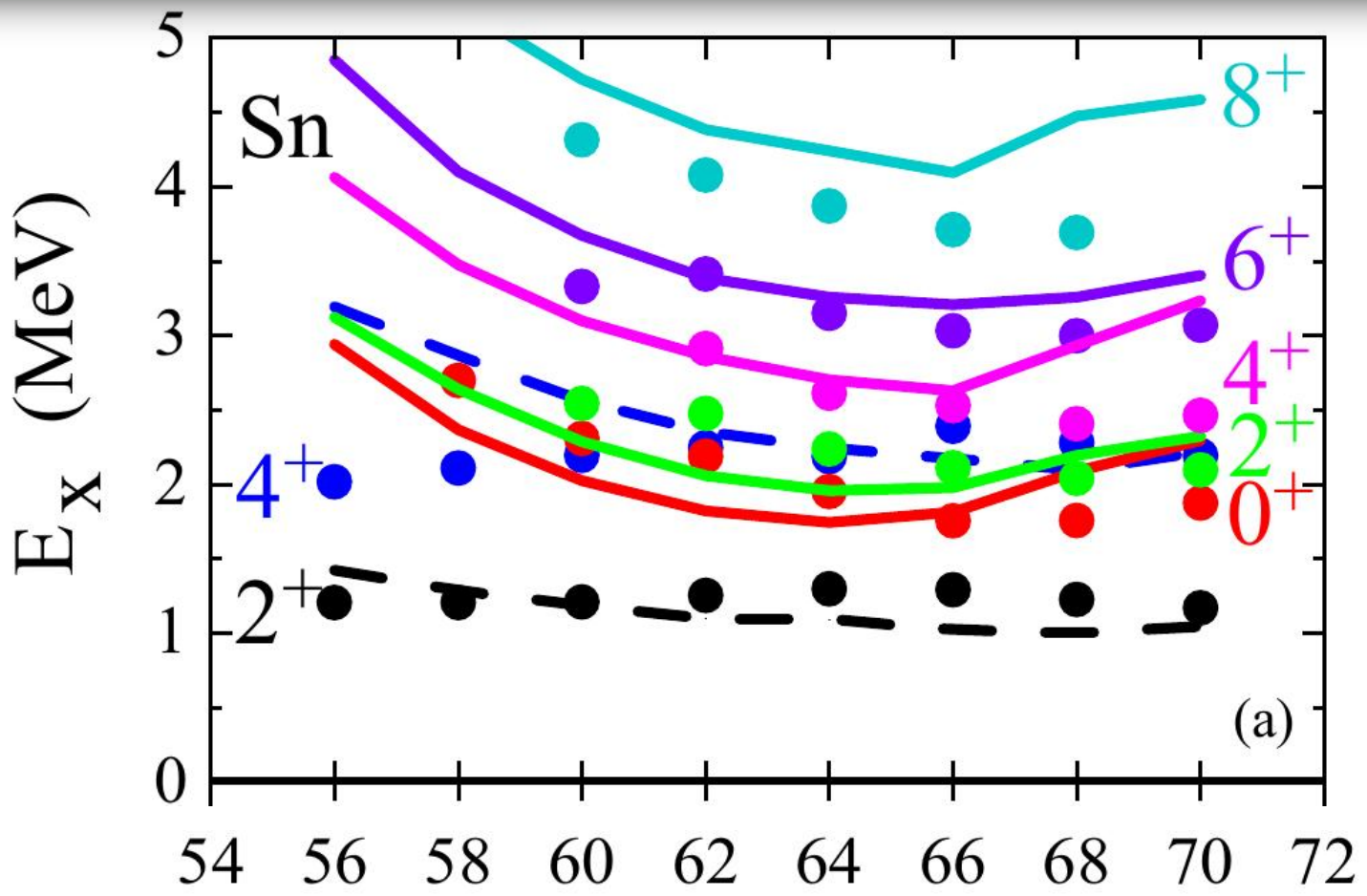
Electromagnetic operators

Effective charges: e_p = 1.2e and e_n = 0.4e. Free-nucleon spin *g* factors are quenched by 0.56.

Interpretation

nex ≈ 2 is an occupation-number expectation value, not the probability of a pure 2p–2h component. Shape assignments use Qs together with projected PESs and collective distributions.

Intruder-band systematics in $^{106-120}$Sn



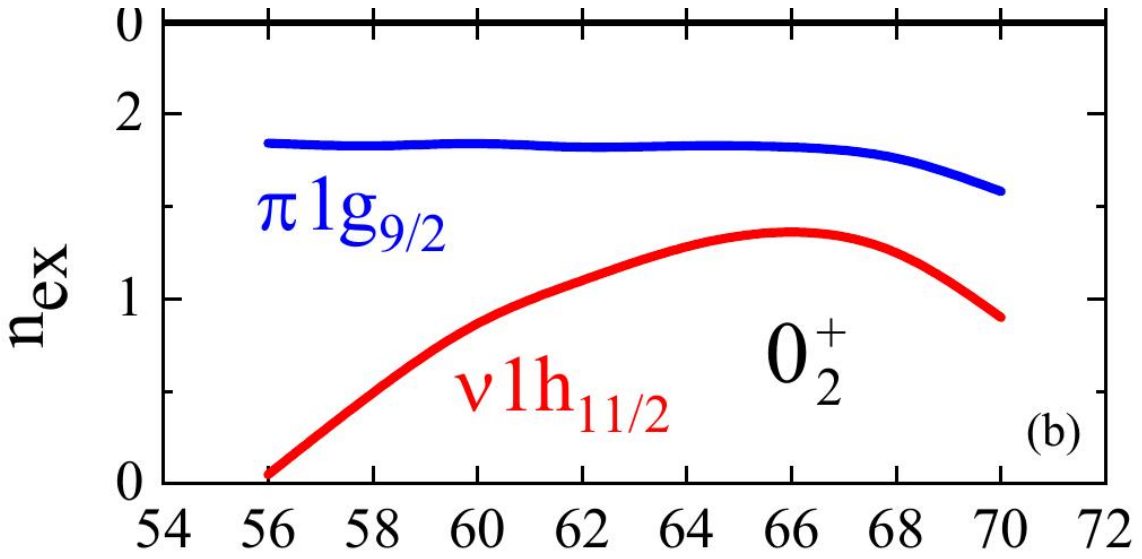
The calculated intruder 0^+–8^+ sequence reproduces the parabolic trend with a minimum at $N = 66$, while the normal 2^+ and 4^+ levels vary only weakly. The experimental points broadly follow the same systematics.

Origin of the parabolic intruder-band trend

Proton 2p–2h and neutron $\nu 1h_{11/2}$ correlations

- $n_{ex}(0_2^+, \pi 1g_{9/2})$ is close to 2 in $^{112-118}$Sn, supporting the cross-shell proton 2p–2h assignment.
- Its isotope dependence is nearly flat and cannot by itself produce the parabolic energy trend.

- $n_{ex}(0_2^+, \nu 1h_{11/2})$ has a bell-shaped dependence with a maximum at $N = 66$.
- Beyond midshell, increasing $\nu 1h_{11/2}$ occupation Pauli-blocks further promotion from lower orbits.



Spectra and configuration mixing in $^{112-118}$Sn

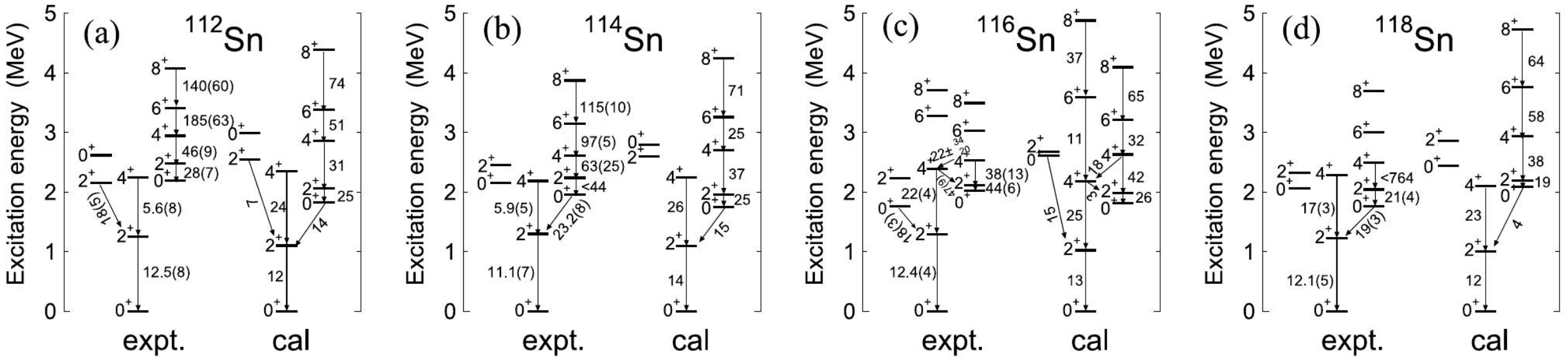


FIG. 2. Comparison between calculated energy spectra with experiment for the Sn isotopes (a) ^{112}Sn, (b) ^{114}Sn, (c) ^{116}Sn, (d) ^{118}Sn. The numbers next to arrows denote the $B(E2)$ values in W.u. Experimental data are taken from [7–9,31,32,39].

The level schemes and collective *E*2 branches are reproduced. Strong 0^+ and 4^+ mixing is found in ^{114,116}Sn, whereas ^{118}Sn is less mixed. In ^{116}Sn the experiment proposed 0_3^+ as the intruder bandhead, while the calculation assigns the dominant intruder character to 0_2^+.

B(E2) values: agreement and remaining deviations

Effective charges: e_p = 1.2e, e_n = 0.4e

TABLE I. Calculated $B(E2)$ values are compared with the experimental data and the *sd* IBM-2 results taken from Refs. [7–9] for ^{114}Sn, ^{116}Sn, and ^{118}Sn.

	^{114}Sn (W.u.)			^{116}Sn (W.u.)			^{118}Sn (W.u.)		
$I_i^+ \rightarrow I_f^+$	Expt.	Calc. IBM		Expt.	Calc. IBM		Expt.	Calc. IBM	
Normal									
$2_1 \rightarrow 0_1$	11.1(7)	14	11	12.4(4)	13	12	12.1(5)	12	10
$4_1 \rightarrow 2_1$	5.9(5)	26	19	22(4)	25	18	17(3)	23	19
Intruder									
$2_2 \rightarrow 0_2$	< 44	25	31	44(6)	26	31	21(4)	19	35
$4_2 \rightarrow 2_2$	62(25)	37	85	38(13)	42	83	< 764	38	39
$6_1 \rightarrow 4_2$	97(5)	25	93		32	90		58	20

- [7] M. Spieker *et al.*, PRC 97, 054319 (2018).
- [8] C. Petrache *et al.*, PRC 99, 024303 (2019).
- [9] K. Ortner *et al.*, PRC 102, 024303 (2020).

For ^{120}Sn:
$B(E2; 2_1 \rightarrow 0_1)_{cal}$ = 10.5 W.u.
$B(E2; 2_1 \rightarrow 0_1)_{exp}$ = 11.41(22) W.u.
While slightly underestimated for $^{106\text{-}110}$Sn

The shell model describes well the systematics of the spectra and *B(E2)* values.

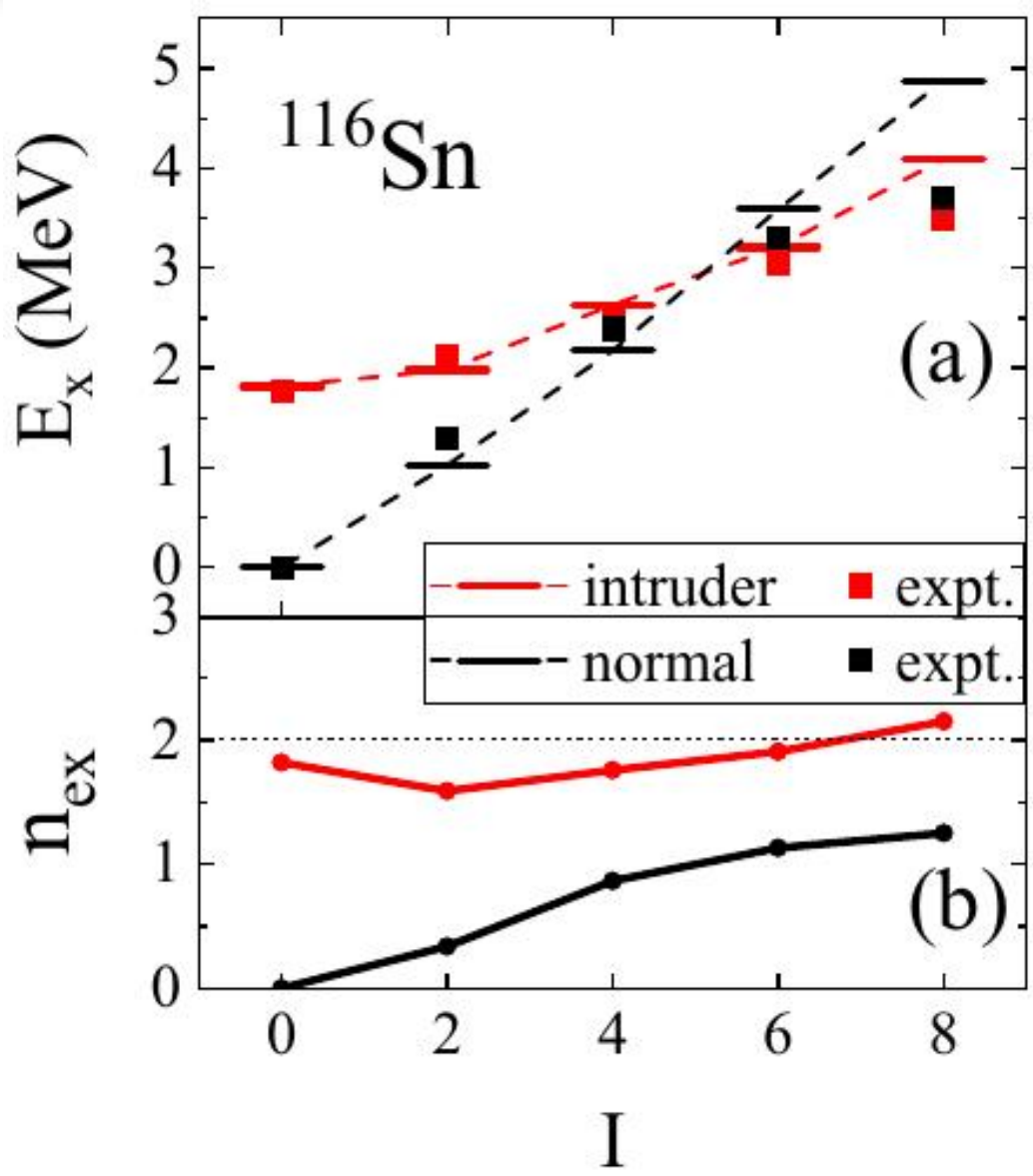
Band crossing and proton excitation in ^{116} Sn

The normal and intruder sequences cross between I = 4 and I = 6

- I ≤ 4: the normal sequence is lower.
- I ≥ 6: the intruder sequence becomes yrast.

- Intruder states retain $n_{ex}(\pi 1g_{9/2}) \approx 2$ from I = 0 to 8.
- The proton excitation is shared 4_2^+ by and 4_1^+, consistent with strong mixing near the crossing.

The crossing exchanges the energy order and dominant band character.



Quadrupole moments and g factors in ^{116}Sn

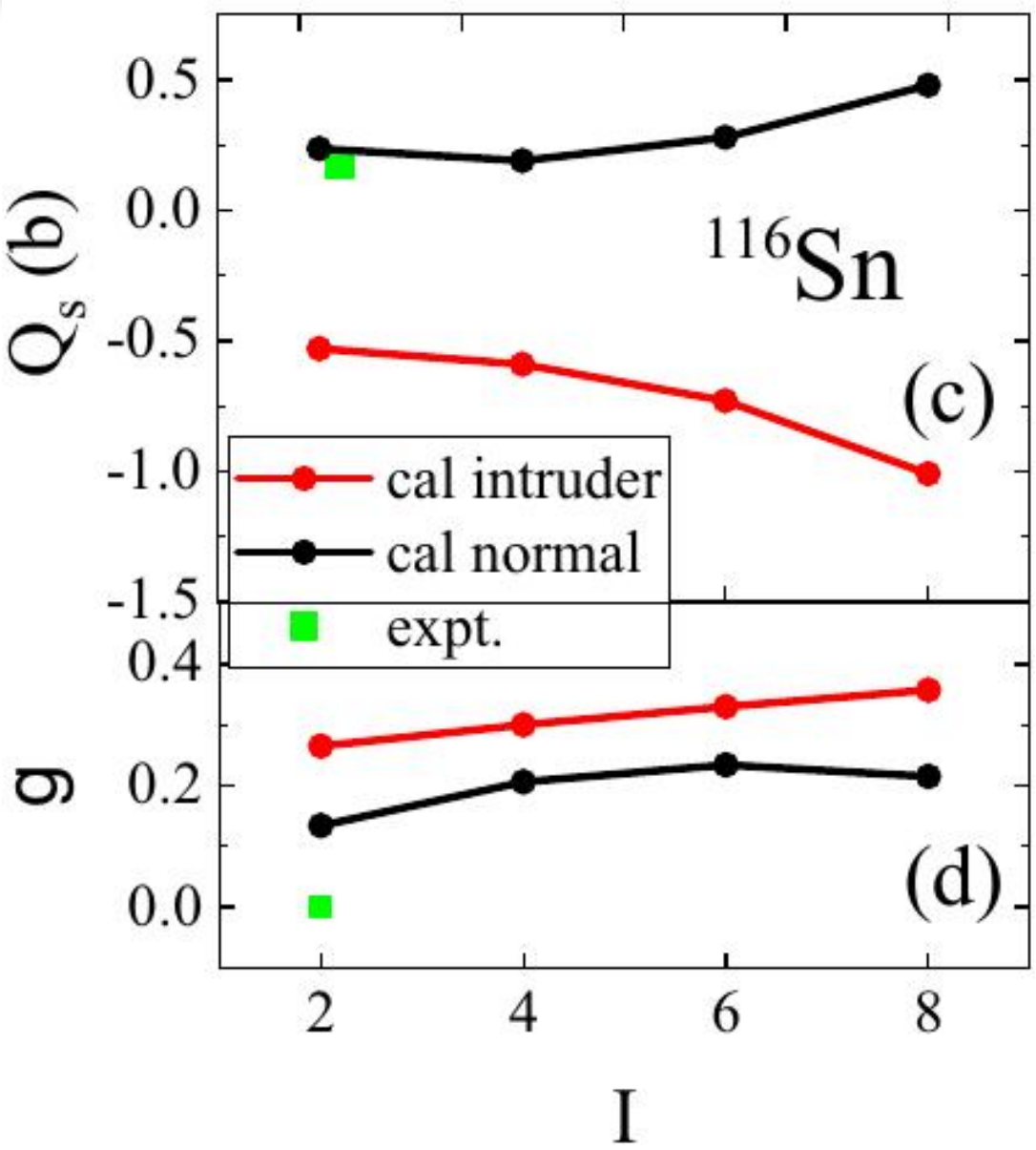
$$Q_s(I, K=0) = -\frac{I}{2I+3} Q_0,$$

$$Q_s < 0 \Rightarrow \text{prolate},$$

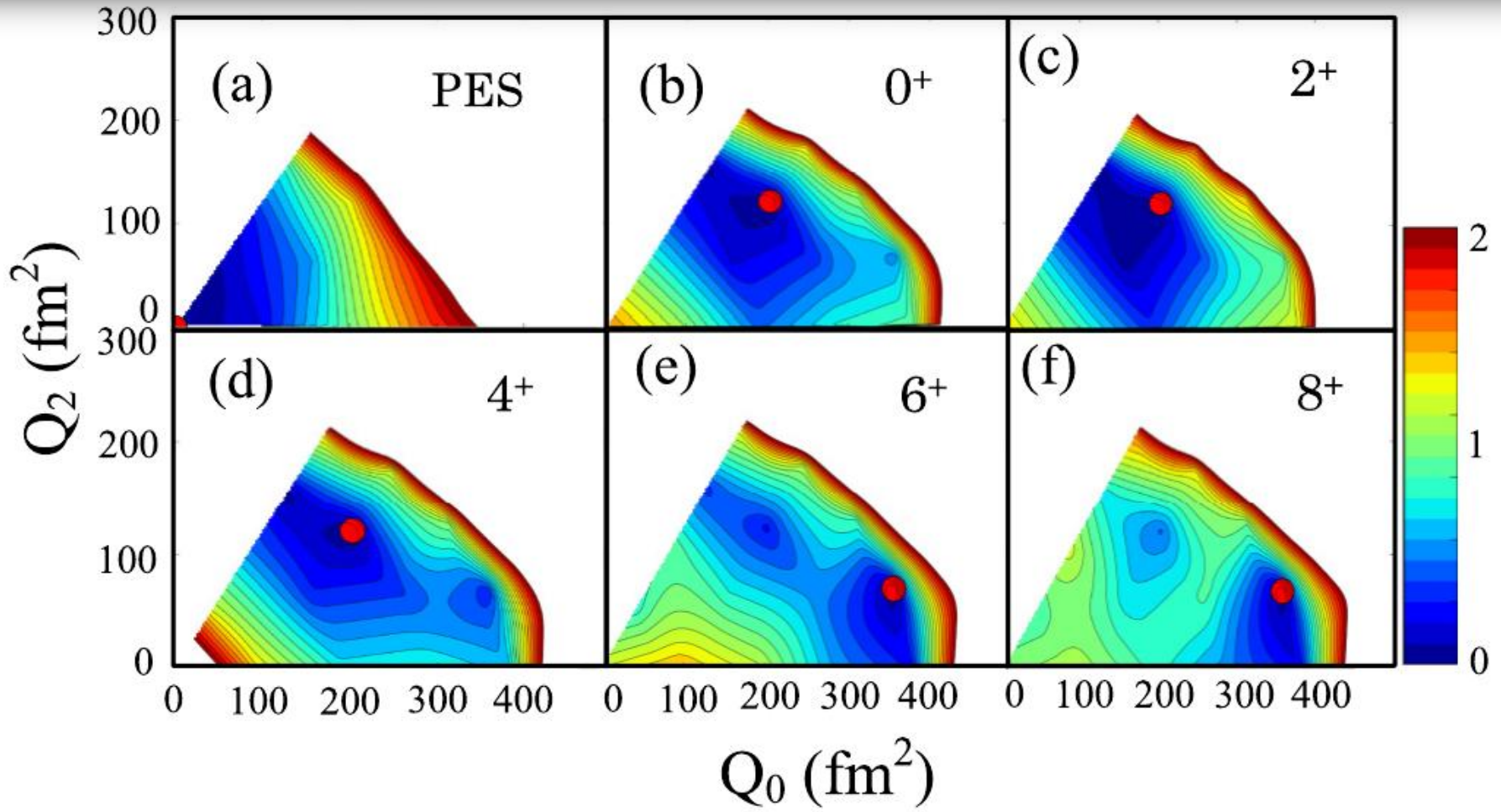
$$Q_s > 0 \Rightarrow \text{oblate}.$$

- Normal states: $Q_s > 0$, oblate-like in the K = 0 rotor convention.
- Intruder states: $Q_s < 0$, prolate-like.
- The calculated $Q_s(2_1^+)$ agrees with the measurement.

- Calculated g factors rise mildly with spin for both sequences.
- With g_s quenched by 0.56, the g factor of 2_1^+ state is still overestimated.

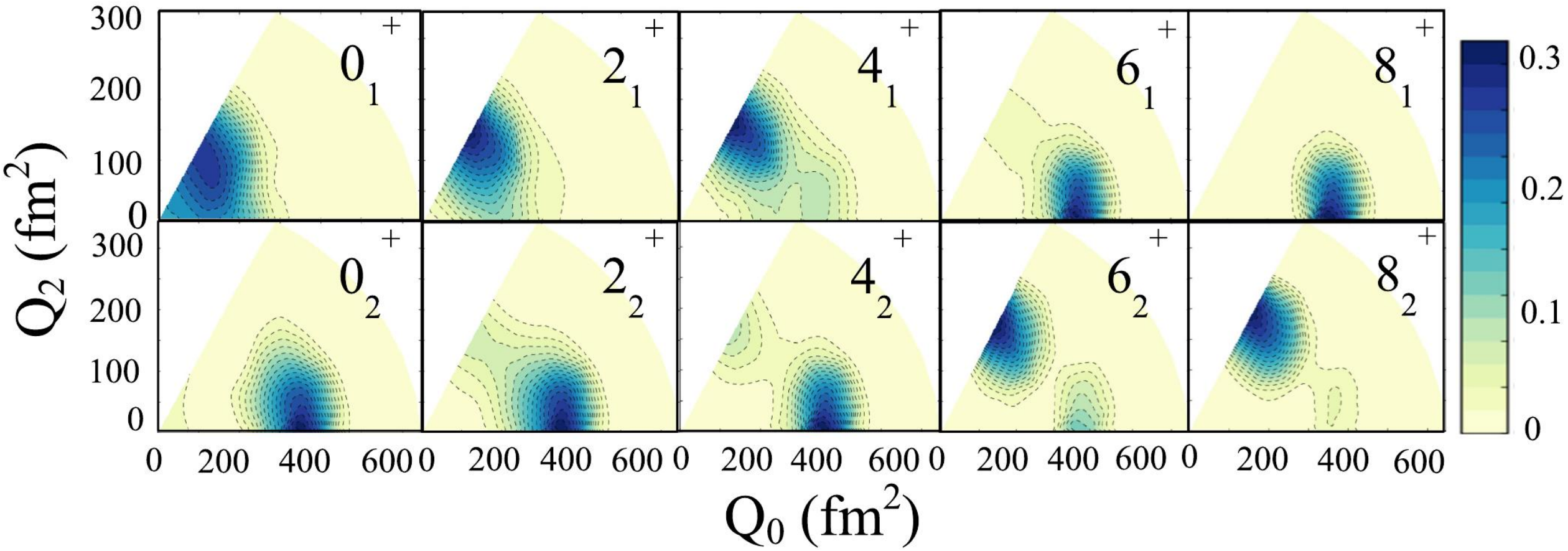


Projected PES in ^{116}Sn



Particle-number projection is retained in every panel. The angular-momentum-unprojected surface has a spherical minimum. After projection, the $I = 0$ and $I = 4$ minima lie on the triaxial/oblate side; a prolate local minimum appears at $I = 4$ and becomes the global minimum at $I = 6$ and 8, while a shallow oblate minimum remains.

Collective wave functions and band-character exchange



For the first states, 0^+_1, 2^+_1 and 4^+_1 are concentrated on the oblate side, whereas 6^+_1 and 8^+_1 move to the prolate side. The second states show the complementary branch; both 4^+_1 and 4^+_2 distributions spread on both sides.

Conclusions and outlook

Microscopic description of intruder states

- The prolate intruder band is generated mainly by proton $2p$–$2h$ excitations across $Z = 50$. Neutron excitation into $\nu 1h_{11/2}$, enhanced near $N = 66$, accounts for the parabolic isotope dependence.

Intruder-nornal band crossing in ^{116}Sn

- The normal and intruder sequences cross between $I = 4$ and $I = 6$; the yrast structure changes from oblate-like to prolate-like.

Limitation and Outlook

- Measure the mixed-state transitions, spectroscopic Q, and g factors in $^{114\text{-}118}$Sn.
- Effective charges and g_s quenching are phenomenological.